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DATA CENTER ENVIRONMENTAL & POWER SOP: THERMAL MANAGEMENT

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1. PURPOSE & HARDWARE CONSTRAINTS

To prevent hardware degradation during high-density local AI inference workloads running on air-gapped server racks.

2. THERMAL THRESHOLDS

- Optimal Operating Inlet Temperature: 18°C to 24°C.
- Thermal Warning Threshold: 28°C ambient temperature.
- Thermal Critical Threshold: 32°C ambient temperature sustained for > 5 minutes.

3. COMPUTATIONAL POWER MITIGATION FORMULA

When ambient rack temperatures exceed 28°C, compute capacity must be throttled down according to:

- $\text{Throttle_Percentage} = (\text{Current_Temp_C} - 28) * 12.5$
- Example: At 32°C, $\text{Throttle_Percentage} = (32 - 28) * 12.5 = 50\%$ workload reduction.

4. SAFETY INTERLOCK

If rack temperature exceeds 35°C, trigger immediate emergency shutdown of non-essential AI compute nodes.